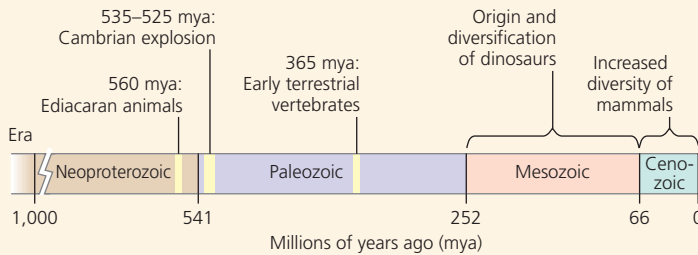


CONCEPT 32.2

The history of animals spans more than half a billion years (pp. 675–679)

- Fossil biochemical evidence and molecular clock analyses indicate that animals arose over 700 million years ago.
- Genomic analyses suggest that key steps in the origin of animals involved new ways of using proteins that were encoded by genes found in choanoflagellates.



? What caused the Cambrian explosion? Describe current hypotheses.

CONCEPT 32.3

Animals can be characterized by body plans (pp. 679–682)

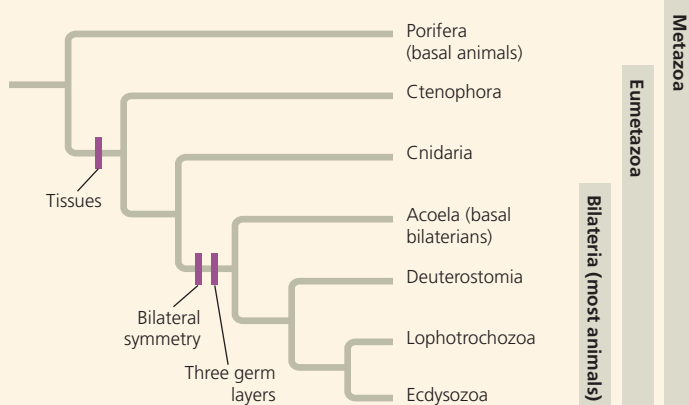
- Animals may lack symmetry or may have **radial** or **bilateral symmetry**. Bilaterally symmetrical animals have **dorsal** and **ventral** sides, as well as **head** and **tail** ends.
- Eumetazoan embryos may be **diploblastic** (two germ layers) or **triploblastic** (three germ layers). Triploblastic animals with a **body cavity** may have a **coelom** or a **hemocoel** (or both).
- Protostome** and **deuterostome development** often differ in patterns of **cleavage**, coelom formation, and **blastopore** fate.

? Describe how body plans provide useful information yet should be interpreted cautiously as evidence of evolutionary relationships.

CONCEPT 32.4

Views of animal phylogeny continue to be shaped by new molecular and morphological data (pp. 682–684)

- This phylogenetic tree shows key steps in animal evolution:



? Consider clades Bilateria, Lophotrochozoa, Metazoa, Chordata, Ecdysozoa, Eumetazoa, and Deuterostomia. List the clades to which humans belong in order from the most to the least inclusive clade.

TEST YOUR UNDERSTANDING

➔ For more multiple-choice questions, go to the **Practice Test** in the **Mastering Biology** eText or Study Area, or go to goo.gl/GruWRg.

Levels 1-2: Remembering/Understanding

- One of the characteristics unique to animals is
 - gastrulation.
 - multicellularity.
 - sexual reproduction.
 - flagellated sperm.
- The distinction between sponges and other animal phyla is based mainly on the absence versus the presence of
 - a body cavity.
 - a complete digestive tract.
 - mesoderm.
 - tissues.
- Which of the following was probably an important factor in bringing about the Cambrian explosion?
 - the movement of animals onto land
 - an increase in the concentration of atmospheric nitrogen
 - the emergence of predator-prey relationships
 - the origin of bilaterian animals

Levels 3-4: Applying/Analyzing

- Based on the tree in Figure 32.11, which statement is true?
 - The animal kingdom is not monophyletic.
 - The Acoela are more closely related to echinoderms than to annelids.
 - Sponges are basal animals.
 - Bilaterians do not form a clade.

Levels 5-6: Evaluating/Creating

- EVOLUTION CONNECTION** A professor begins a lecture on animal phylogeny (as shown in Figure 32.11) by saying, “We are all worms.” In this context, what did she mean?
- WRITE ABOUT A THEME: INTERACTIONS** Animal life changed greatly during the Cambrian explosion, with some groups expanding in diversity and others declining. Write a short essay (100–150 words) interpreting these events as feedback regulation at the level of the biological community.
- SYNTHESIZE YOUR KNOWLEDGE**



This organism is an animal. What can you infer about its body structure and lifestyle (that might not be obvious from its appearance)? This animal has a protostome developmental pattern and a trochophore larva. Identify the major clades that this animal belongs to. Explain your selection, and describe when these clades originated and how they are related to one another.

For selected answers, see Appendix A.